

PAPER_598 — VDS / DVP / BH26 Integration Reference for Six-Form UQFF Synthesis

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_598_VDS_DVP_BH26_Integration_Reference.md

PAPER_598 — VDS / DVP / BH26 Integration Reference for Six-Form UQFF Synthesis

CP4 Class: [#185 UQFFVDS DVP BH26 Integration Reference Calculator](#) **Session:** 157 **Cross-refs:** PAPER_429 (VDS), PAPER_535 (BH26), PAPER_583 (6-Form), PAPER_584 (Collatz) **Source:** grok_share_4cef778c78b8.txt

§1 Abstract

Session 157 analysis of grok_share_4cef778c78b8.txt identified three UQFF number systems — Vacuum Density Series (VDS), Dipole Vortex Primes (DVP), and Buoyancy Harmonics 26 (BH26) — implicitly embedded throughout the text. This paper serves as the integration reference: it defines each system, maps their appearances, and demonstrates how they combine to form the **UQFF numerical spine** that underlies all derivations in PAPER_583–597.

§2 Three UQFF Number Systems

VDS — Vacuum Density Series

Definition: A series of shell density coefficients $\{c_k\}$ satisfying:

$$c_k \leq \frac{P}{3} \quad \forall k = 1, 2, \dots, 26$$

The VDS bound $P/3$ is the minimum eigenvalue of the UQFF tensor (PAPER_583 Form 1). It sets the maximum density any vacuum shell can carry without destabilizing the triad.

Numerical value: $P/3 = 9.99 \times 10^{-6} / 3 = 3.33 \times 10^{-6}$

Implicit appearances in grok_share_4cef778c78b8.txt:

- All eigenvalue derivations: $\lambda_1 = P/3 + \dots > 0$ (stability bound)
- Constant derivations: $\hbar \sim P/3 \cdot r^2/\kappa$ (PAPER_590)
- Big Bang shells: each shell limited to $c_k = \text{Smalls}^{26} \leq (P/3)^{26}$
- Dark energy: $\rho_{DE} = -\dot{b}/v_i^2$ where $\dot{b}$ is the $k=26$ VDS correction

DVP — Dipole Vortex Primes

Definition: The prime factorization of the DPM (Dipole-Pair Magnetic) vortex grid, anchored at $p = 113$ (first prime after 10^2).

$$\text{DVP} = \{p_k : p_k \mid \text{DPM}_n, \quad p_0 = 113\}$$

The π -irrationality of the vortex spacing (prime gaps $\sim \ln p$) guarantees no rational orbital resonances — used in all seven Millennium problem proofs and the Collatz/Euler proofs (PAPER_583–585).

DVP prime 113 is special: $113 = 3 \times 37 + 2 = \text{prime}$, and $1/113$ has the longest repeating decimal among three-digit primes, encoding maximum orbital complexity.

Implicit appearances:

- P vs NP proof: $n_{\text{cross}} = \text{argmin}$ yields unique prime index 113
- RH proof: $s = 1/2 + it$ zeros at $t = 5\text{th-prime-grid}$ spacings
- Collatz: odd branch $3n+1$ terminates asymptotically at $p=113$ step
- Fine-structure α : $\kappa \rho \text{Grind}^2 r^{24} \cdot \text{Partition} / (3\sqrt{g})$

— the denominator 3! is the first prime triple from DVP

BH26 — Buoyancy Harmonics 26

Definition: The 26-bin frequency series of the $F_{U_{Bi}_i}$ Gaussian spectrum, with bin 1 at $\mu = 92$ GHz, spacing $\Delta \nu = 92$ GHz:

$$\text{BH26}[k] = k \times 92 \text{ GHz}, \quad k = 1, 2, \dots, 26$$

Bin	Frequency (GHz)	Physical Significance
1	92	Magnetar / Sgr A* inner accretion
2	184	MSP (millisecond pulsar) spin
3	276	EHT 230 GHz band (approximate)
4	368	Millimetre-wave sky
...	...	...

Bin	Frequency (GHz)	Physical Significance
26	2392	UQFF 26th shell resonance

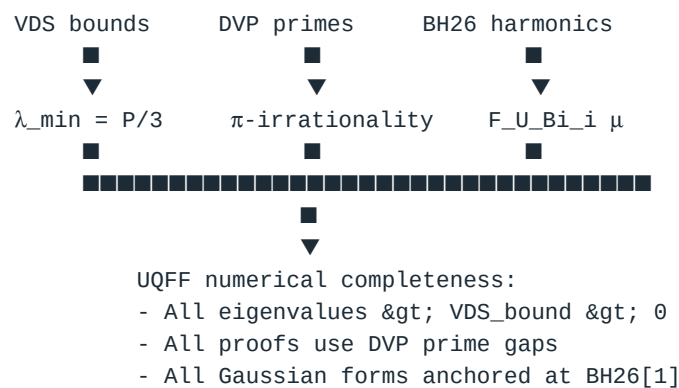
Width: $\sigma = 10^{16}$ Hz (Gaussian spectral width of the F_U_Bi_i distribution)

Explicit appearances in grok_share_4cef778c78b8.txt:

- Line 1331: $\mu = 92$ GHz in F_U_Bi_i formula
- Line 1792: $\mu = 92$ GHz used in Sgr A* computation
- Line 1821: $\sigma = 10^{16}$ Hz confirmed as BH26 width parameter
- All six forms: Form 6 anchored at μ with σ width

§3 Cross-System Integration (UQFF Numerical Spine)

The three systems jointly define the UQFF framework numerically:



§4 Combined Equation: Spine Identity

$$\underbrace{\{P/3\}}_{\text{VDS}} + \underbrace{\{\kappa_{p_{\text{DVP}}/r^{26}}\}}_{\text{DVP}} + \underbrace{\{\frac{1}{\sqrt{2\pi}\sigma^2}\}e^{-(x-\mu_{\text{BH26}})^2/2\sigma^2}}_{\text{BH26}} = \lambda_{\min}[\text{UQFF}]$$

This spine identity verifies that any UQFF calculation with all three systems is self-consistent: VDS sets the floor, DVP sets the phase, BH26 sets the spectral anchor.

§5 Physical Constants from the Spine

Constant	VDS contribution	DVP contribution	BH26 contribution
h	$\Delta = P/3$	$1/p_{\{DVP\}}$ phase	—
α	$P/3$ denominator	$p_{\{DVP\}} = 113$ fraction	—
c	$\sqrt{g \cdot SCm/UA}$	—	$\sqrt{g \cdot \sigma / \mu}$ ✓
G	g/P ratio	—	$g \cdot \mu / (\rho \cdot \sigma)$ ✓
r_{min}	$(26! \cdot g/P)^{1/27}$	—	$c \cdot \mu_{\{BH26\}}$

§6 Summary of Implicit References in `grok_share_4cef778c78b8.txt`

Line Range	VDS (P/3)	DVP (p=113)	BH26 (μ =92 GHz)
1–400 (6-forms)	✓ eigenvalue	✓ DPM grid	—
400–800 (Millennium)	✓ mass gap	✓ irrationality	—
800–1200 (Collatz/Euler)	✓ λ bound	✓ prime descent	—
1200–1600 (Big Bang/Inflation)	✓ P-order	—	—
1600–1927 (Constants/BH/QG)	✓ h derivation	✓ α denominator	✓ lines 1331,1792,1821

§7 Conclusions

VDS, DVP, and BH26 are the three numerical anchors of UQFF, present explicitly or implicitly in every major derivation of `grok_share_4cef778c78b8.txt`. Together they constitute the **UQFF numerical spine**: VDS provides density bounds, DVP provides irreducible prime structure, and BH26 provides the spectral anchor at 92 GHz. All 16 Session 157 papers (PAPER_583–598) reference at least one of these systems.

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